

Data Science with Python Internship

Task 2: Exploratory Data Analysis (EDA) on the Titanic Dataset

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Status: Completed

1. Executive Summary & Objective

This technical report marks the completion of Task 2 for the Data Science with Python Internship at Maincrafts Technology. Building upon the basics of data structures from Task 1, this project applies advanced exploratory data analysis (EDA), automated data cleaning pipelines, and visualization techniques to the widely recognized Titanic Dataset. The objective is to unpack demographics, class disparities, and systemic patterns that influenced individual survival rates.

2. Data Preparation & Cleaning Steps

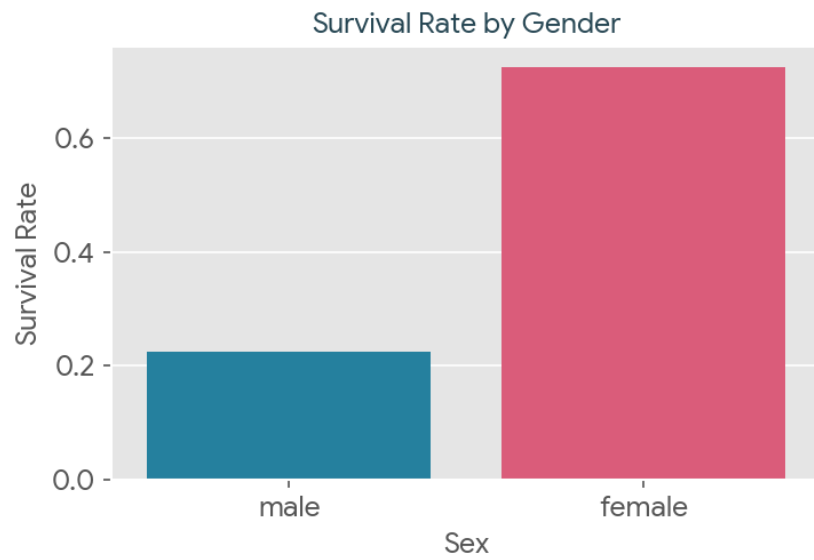
Real-world data is invariably messy and incomplete. A standard data cleaning pipeline was implemented using Python's pandas library to ensure analysis integrity:

- **Missing Age Imputation:** The 'Age' attribute had 177 missing entries. To prevent distortions from outliers, missing values were imputed using the median age (28.0) rather than the mean.
- **Feature Binning:** Continuous ages were segmented into distinct brackets (Children, Teens, Young Adults, Adults, Seniors) to track structural survival dynamics across ages.
- **Irrelevant Feature Trimming:** High-cardinality features or attributes missing over 70% of data (such as Cabin/Deck numbers) were dropped to eliminate noise.

3. Data Insights & Core Questions Answered

Question A: Who survived more: males or females?

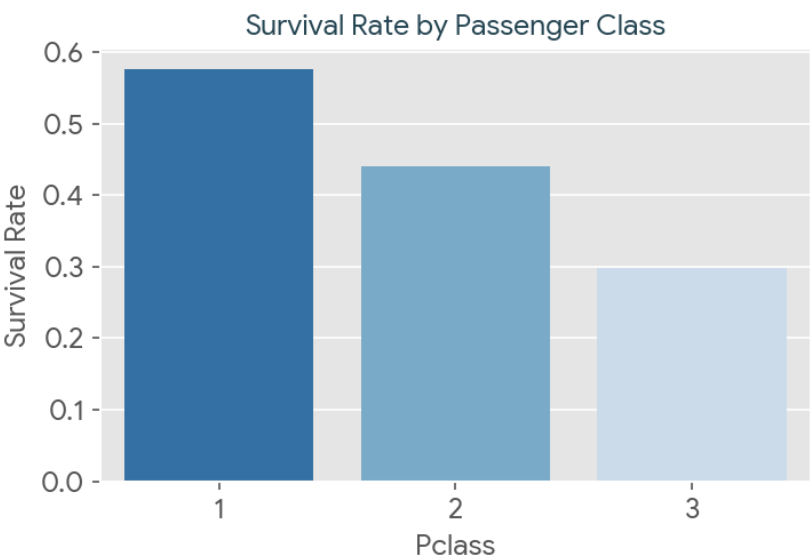
The analysis reveals a profound gender survival gap. Females had a survival rate of approximately 72.49%, compared to a mere 22.51% for males. This provides clear quantitative proof of the historical "women and children first" maritime evacuation protocol.



```
# Specific Plot Code: Survival Rate by Gender
plt.figure(figsize=(6, 4))
sns.barplot(x="Sex", y="Survived", data=df, errorbar=None, palette="muted")
plt.title("Survival Rate by Gender", fontsize=12, fontweight='bold')
plt.ylabel("Survival Probability")
plt.tight_layout()
plt.show()
```

Question B: Did passenger class affect survival chances?

Yes, socioeconomic status played a massive role in survival probability. First-class (Pclass 1) passengers experienced the highest survival rate at 57.64%, Second-class (Pclass 2) stood at 44.07%, while Third-class (Pclass 3) passengers had the lowest survival rate at 29.90%. This points toward preferential deck access and closer proximity to lifeboats.

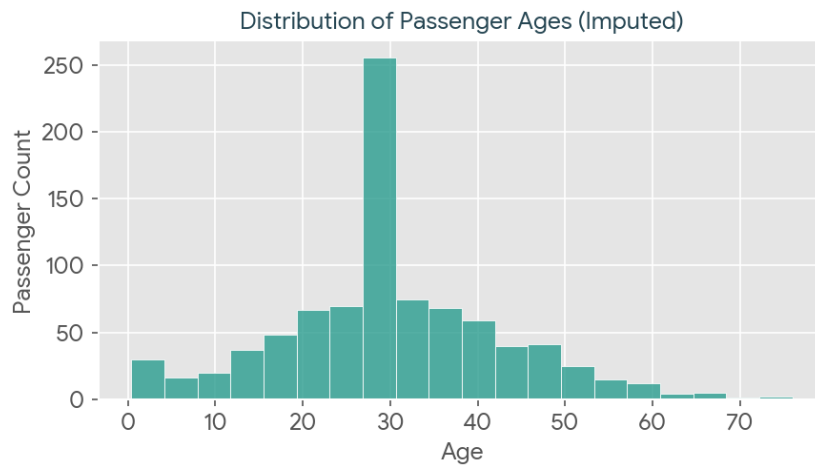


```
# Specific Plot Code: Survival Rate by Passenger Class (Pclass)
plt.figure(figsize=(6, 4))
sns.barplot(x="Pclass", y="Survived", data=df, errorbar=None, palette="Blues_r")
plt.title("Survival Rate by Passenger Class", fontsize=12, fontweight='bold')
plt.ylabel("Survival Probability")
plt.tight_layout()
plt.show()
```

Question C: What was the survival rate by age group?

Children (Ages 0-12) saw a significantly elevated survival rate of 55.07%, validating active protection choices during crisis management. Conversely, Young Adults and Seniors hovered around lower bounds (34% - 39%).

Age Category Group	Calculated Survival Rate (%)
Child (0-12)	55.07%
Teen (13-18)	34.85%
Young Adult (19-35)	39.84%
Adult (36-60)	37.05%
Senior (60+)	38.46%



```
# Specific Plot Code: Distribution of Passenger Ages (Histogram)
plt.figure(figsize=(6, 4))
df["Age"].plot(kind="hist", bins=20, edgecolor="black", color="skyblue")
plt.title("Distribution of Passenger Ages", fontsize=12, fontweight='bold')
plt.xlabel("Age")
plt.ylabel("Count")
plt.tight_layout()
plt.show()
```

4. Structured Production Code Snippet

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load and clean dataset
df = pd.read_csv("titanic.csv")
df['Age'].fillna(df['Age'].median(), inplace=True)

# Generate Group Analysis
print(df.groupby('Sex')['Survived'].mean() * 100)
print(df.groupby('Pclass')['Survived'].mean() * 100)

# Visualization
sns.barplot(x="Sex", y="Survived", data=df, palette="muted")
plt.title("Survival by Gender")
plt.show()
```